

Numerical Methods: 24 Hours Take-home examination, June 2026

This examination consists of 5 exercises. Exercises 1–4 are for all students. Exercise 5a is for students taking the 5 ECTS version ONLY, and Exercise 5b is ONLY for students taking the 7 ECTS version. If you answer the wrong exercise of 5a and 5b, you will get zero points for that. Each question has been assigned a weight (points). The total number of points is 100.

For the exam, you will need to be able to read matrices and vectors from files having the following format for an $M \times N$ matrix:

M
 N
 $Row1$
 $\dots$
 $RowM$

where vectors always have $N = 1$. An example with a 3×2 matrix could be

3
2
3.04464 5.06464
−0.6454 0.435435
4.05454 −1222.933435

You must hand in a zip file containing a report and all your used code. Concerning the report, it needs to be CLEARLY readable (unreadable parts will be assessed as wrong). Whenever the word "state" is used in the questions, the answer MUST be present in the report. Wherever it says "Submit the used code", the used code MUST be handed in. Otherwise, you will get zero points for your answer. Clearly name your code files so that it is easy to see what exercise the code is used for.

Notice the following rules:

- It is allowed to use general purpose methods, text or code that are part of NR or elsewhere publicly available, including methods, text or code that has been uploaded by Peter or me during the course. However, there must for each use be a CLEAR marking **stated in the report** including where the methods/text/used code was taken from. In all cases, you are solely responsible for the correctness of used methods, text and code. All methods, text and code that **explicitly handles the problems from the exam exercise** MUST be written by yourself.
- It is NOT allowed to share your answers (or parts of these), or to communicate with other people about the exercise. This includes other NM students taking this 24 hours take-home examination.

I wish you all the best for the forthcoming 24 hours!

Best regards, Henrik

Exercise 1 (30 points)

Consider a Linear Least Squares problem with a 48×10 design matrix $\mathbf{A}$ and the right hand side $\mathbf{b}$ that are given in *Ex1A.dat* and *Ex1b.dat* respectively.

- i) (5 points) Find the Singular Value Decomposition $\mathbf{A} = \mathbf{U}\mathbf{W}\mathbf{V}^T$. State the diagonal elements in $\mathbf{W}$. Submit the used code.

The SVD method automatically removes the singularities to a user specified level when computing the solution by deploying the well-known threshold. The smallest diagonal element in $\mathbf{W}$ is very close to zero

- ii) (5 points) Use the Singular Value Decomposition and set the threshold to compute the solution $\mathbf{x}$ to $\mathbf{Ax} = \mathbf{b}$ where only the solution components related to the smallest element in $\mathbf{W}$ are removed. State the solution $\mathbf{x}$ and the residual error $\|\mathbf{Ax} - \mathbf{b}\|$. Submit the used code.

The second and third smallest element are also very small.

- iii) (5 points) Use the Singular Value Decomposition and set the threshold to compute the solution $\mathbf{x}$ to $\mathbf{Ax} = \mathbf{b}$ where the solution components related to all three very small elements in $\mathbf{W}$ are removed. State the solution $\mathbf{x}$ and the residual error $\|\mathbf{Ax} - \mathbf{b}\|$. Submit the used code.
- iv) (5 points) It can be seen that the residual error in ii) is a little bit smaller than the residual error in iii). State an explanation for this.
- v) (5 points) State an estimate on the error on the solution $\mathbf{x}$ to the problem from both ii) and iii). State an explanation for how these estimate were obtained. Also state a comparison discussion between the two results. Submit the used code
- vi) (5 points) Use the information from the SVD matrices to state an orthonormal basis for the set of linear combinations of elements in $\mathbf{x}$ that were excluded from the solution in question iii).

Exercise 2 (15 points)

Consider the equation

$$f(x) \equiv 3x \cos^3(x) - 0.2 \exp(x^2) + 0.7 = 0$$

- i) (2 points) State (with at least 7 digits) the value of $f(-1)$ and $f(1)$. (HINT: you should get something around $f(-1) \simeq -0.32$, $f(1) \simeq 0.63$)
- ii) (5 points) Search for a solution to the equation using 4 iterations of Ridders method with the initial bracket $x_0 = -1, y_0 = 1$. State for each iteration i the values of x_i, y_i . Submit the used code.
- iii) (5 points) State an estimate of the accuracy after 5 iterations. The estimate must be based only on the data obtained from the 5 iterations and must be stated with a clear argument of how you computed it. Without such a clear argument, there is no points for the answer.

As mentioned in the course, Ridders method is both robust and fast.

- iv) (3 points) State what is the expected "order per f-computation" for Ridders method. State also (with a clear argument) what may lead to a smaller actual than expected order.

Exercise 3 (20 points)

Consider an inverted planar pendulum with a vibrating suspension point. The movement of the suspension point p is given as

$$p(t) = (0, A \sin(\omega t))$$

The pendulum rod has length L and can be viewed as massless. The mass of the pendulum is m . Let $\theta(t)$ be the angle of the pendulum so that the location of the pendulum is given as

$$\begin{aligned} r(t) &= p(t) + L(\cos(\theta(t)), \sin(\theta(t))) \\ &= (L \cos(\theta(t)), A \sin(\omega t) + L \sin(\theta(t))) \end{aligned}$$

We then obtain

$$\dot{r}(t) = (-L \sin(\theta(t))\dot{\theta}(t), A\omega \cos(\omega t) + L \cos(\theta(t))\dot{\theta}(t))$$

and hence the kinetic energy is

$$T = \frac{1}{2} m \dot{r}(t)^2 = \frac{1}{2} m L^2 \dot{\theta}(t)^2 + A L \omega \cos(\omega t) \cos(\theta(t)) \dot{\theta}(t) + \frac{1}{2} A^2 \omega^2 \cos^2(\omega t)$$

With the gravitational acceleration being $(0, -g)$, we obtain the potential energy

$$V = mg(A \sin(\omega t) + L \sin(\theta(t)))$$

We then obtain the Euler-Lagrange equation of motion for θ as

$$m L^2 \ddot{\theta}(t) - A L \omega^2 \sin(\omega t) \cos(\theta(t)) - mg L \cos(\theta(t)) = 0$$

We reorganize this equation and add air resistance:

$$\ddot{\theta}(t) = \left[-\frac{g}{L} + \frac{A \omega^2}{m L} \sin(\omega t) \right] \cos(\theta(t)) - k_D \dot{\theta} |\dot{\theta}| \quad (1)$$

- i) (4 points) Rewrite Eq.(1) into a set of two first order ODE's. State the two first order ODE's including the initial conditions.

We have the parameters $m = 1$, $L = 0.5$, $g = 9.81$ and $k_D = 0.2$. Using approximations, it can be shown that the inverted pendulum is stable if $\omega \gg \sqrt{\frac{g}{L}}$ and $A\omega > \sqrt{2gL}$. Hence, we choose $\omega = 100$ and $A = 0.05$.

We wish to numerically estimate $\theta(t)$ for $t \geq 10$. We select $\theta(10) = 1.1$; $\dot{\theta}(10) = -0.1$.

- ii) (3 points) State with at least 10 digits the value of $\ddot{\theta}(10)$. Submit the used code. HINT: You should get $\ddot{\theta}(10) \simeq 366.2$.
- iii) (6 points) Use the Midpoint method with $N = 100000$; $h = 0.0001$ to generate a solution for $\theta(t)$; $10 \leq t \leq 20$. State the value of $\theta(t)$ at $t = 20$. State a plot of $\theta(t)$; $10 \leq t \leq 20$. Submit the used code.

You may in the last question assume that the global order is 2 as expected.

- iv) (7 points) Use now the Midpoint Method with the usual approach with subdivisions $N = 1000, 2000, 4000, \dots$; $h = 10.0/N$. Use as few number of subdivisions as possible to reach an estimated accuracy better than 10^{-5} on $\theta(20)$. State a clear explanation of how you estimate the accuracy and at which subdivision you have reached the required accuracy.

Exercise 4 (20 points)

Consider the integral

$$\int_1^4 f(x) dx$$

where $f(x) = \exp(2x)\sqrt{\sin(-x^2 + 5x - 4)}$

We wish to approximate the integral using the Simpson method. For this, we as usual split the integration interval into N equidistant subintervals.

- i) (8 points) With $N-1 = 2^k$; $k = 1, \dots, 20$ use the Simpson Method method to approximate the integral. State the results in a table similar to those used during the course. Submit the used code.
- ii) (4 points) Use Richardson extrapolation to estimate the order at $N-1 = 2^{20}$. State the result. Submit the used code.
- iii) (4 points) The estimated order is different than the expected order. State an explanation for the difference and your guess on the exact order that you have estimated.
- iv) (4 points) State the estimated accuracy on the result at $N-1 = 2^{20}$. State clearly how you compute the accuracy estimate.

Exercise 5a (15 points). (Only for students taking the 5 ECTS version)

Consider the problem

$$\begin{cases} y''(x) = \cos^3(y(x)) + y'(x)^2 - 0.3 \exp(-x) \cos(y'(x)); & 0 < x < 8 \\ y(0) = -0.6; & y(8) = 0.4 \end{cases}$$

- i) (8 points) Use the Finite Difference method to find an approximation to the solution curve $y(x)$. Use $N = 2, 4, 8, 16, 32, \dots, 32768$. State the numerical estimate of $y(4)$ for each N with at least 10 digits. Submit the used code.
- ii) (7 points) Determine the smallest N at which you can obtain an estimated accuracy on $y(4)$ of less than 10^{-6} . State N , state your estimate of the accuracy, and state an explanation on how you found this estimate.

Exercise 5b (15 points). (Only for students taking the 7 ECTS version)

Consider the problem

$$\begin{aligned} \frac{\partial u(x, t)}{\partial t} &= 2 \frac{\partial^2 u(x, t)}{\partial x^2} + (1 - x) \cos(\pi x) \exp(-\sqrt{t}); & 0 < t; & 0 < x < 1 \\ u(x, 0) &= x^2 + 3; & & 0 \leq x \leq 1 \\ u(0, t) &= 3; & u(1, t) &= 4 + \sin(t); & 0 < t \end{aligned}$$

- i) (10 points) Estimate $u(\frac{1}{2}, 10)$ with a proven accuracy better than 5×10^{-8} using Crank-Nicolson. Start with $\Delta x = \Delta t = \frac{1}{4}$ and perform subdivisions as suggested in the course. State at which subdivision you were able to reach the desired accuracy, your approximation for $u(\frac{1}{2}, 10)$ at that subdivision and your accuracy estimate. State a clear argument for how you arrived at the accuracy estimate. Submit the used code.

Consider now Poissons equation with Dirichlet Boundary conditions on the rectangular box bounded by the corners $(0, 0)$; $(8, 6)$, $(-1.5, 2)$ and $(6.5, 8)$.

- ii) (5 points) State how you would place the grid and with $h = 0.1$ state how many equations you would obtain and state the bandwidth of the resulting matrix.